

WASATCH BIOLABS

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# LIMS Research Report

Wasatch.BioLabs

This research report outlines findings from user research to guide improvements in Wasatch BioLabs' LIMS system. It identifies user needs, pain points, and opportunities to enhance usability, efficiency, and scalability, laying the foundation for design decisions.

## OBJECTIVE

Our objective was to enhance the existing LIMS by improving key features, particularly focusing on user management, sample submission, order tracking, and report generation.

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How can we improve the process of account creation and user management to make onboarding more efficient and user-friendly?

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In what ways can we simplify the sample and order submission workflows to enhance usability and support scalability?

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What changes can be made to real-time tracking, report generation, and results access to improve user satisfaction and optimize system performance?



METHODOLOGY

Timeline

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1 - 2  
weeks

This time frame covers the entire research process from initial stakeholder interviews to final analysis and reporting.

Research Activities

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4  
activities

Analyzed existing competitor solutions and synthesized findings into user stories, journey maps, and user flows to identify opportunities for system improvements.

Data collection

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3  
techniques

Conducted in-depth reviews of existing literature and competitive products, and interviews with key stakeholders to gather essential insights.

## EXECUTIVE SUMMARY

### Theme 1

Our analysis of competitive solutions revealed that streamlined user management systems significantly enhance user engagement and efficiency.

Integrate more intuitive user management features that allow quick modifications and real-time feedback on changes.

### Theme 2

User stories and journey maps identified a need for simpler workflows for sample and order submissions to reduce entry errors and increase throughput.

Develop a unified workflow for sample and order processing that includes bulk operations and simplified steps, enhancing user productivity.

### Theme 3

Feedback from user flows suggests a lack of adequate real-time tracking and customizable reporting capabilities.

Introduce advanced real-time tracking systems and flexible, user-friendly reporting tools that support a range of user needs from basic to advanced analysis.

# Competitive & SWOT Analysis

This section outlines the key strengths, weaknesses, opportunities, and threats identified in our research of leading LIMS systems.

Note: Full Competitive & SWOT analysis and details can be accessed in Figma:

[Link to Competitive Analysis in Figma](#)

## Findings

### Scalability

Competitors struggle with scaling their systems for larger labs, indicating an opportunity for Wasatch to develop features that grow with lab needs.

### Customization and Usability

Many systems are difficult to customize, presenting an opportunity to create more flexible, user-friendly customization options.

### Tracking & Reporting

Barcode and RFID tracking is a strength across competitors but has pain points, like slow searches and limited multi-test access. Competitors struggle with report customization and real-time updates, suggesting a need for intuitive, customizable report generation tools.



# User Stories

User stories capture specific needs and goals from various user perspectives. These stories guide feature development and ensure that the design aligns with real user needs. They are crucial for understanding the motivations behind user interactions.

Note: Full User stories and details can be accessed in Figma:

[Link to User stories in Figma](#)

| User Type      | Goal                                                                    | Outcome                                                        |
|----------------|-------------------------------------------------------------------------|----------------------------------------------------------------|
| Lab Technician | I want to submit samples quickly and easily.                            | So I can ensure timely testing and tracking of the samples.    |
| Admin          | I want to manage user roles and permissions efficiently.                | So the lab runs securely and workflows are uninterrupted.      |
| Researcher     | I want to access my test results immediately.                           | So I can make data-driven decisions in a timely manner.        |
| Admin          | I want to approve or reject new user registrations efficiently.         | So I can ensure only authorized personnel access the LIMS.     |
| Lab Technician | I want to bulk upload sample data.                                      | So I can save time when submitting multiple samples.           |
| Researcher     | I want to generate customizable reports for specific tests.             | So I can easily analyze data and share findings.               |
| Admin          | I want to track the status of submitted samples in real-time.           | So I can manage lab operations effectively and meet deadlines. |
| Lab Technician | I want to receive notifications when sample status changes.             | So I can stay updated and take necessary actions quickly.      |
| Researcher     | I want to compare historical test results across different time points. | So I can identify trends and make informed decisions.          |
| Admin          | I want to assign user roles based on institutional affiliation.         | So the right people have appropriate access to LIMS features.  |
| Researcher     | I want to access past reports easily.                                   | So I can reference them in future research without delays.     |



# User Journey Maps

User journey maps illustrate the step-by-step processes that users go through when interacting with the LIMS. These maps highlight pain points and opportunities for improvement, providing a clear vision of the user experience at different stages.

Note: Full User journey maps and details can be accessed in Figma:

[Link to Journey Maps in Figma](#)

| Feature                            | Key Steps                                                                                                                                               | Outcome                                                                                                       | Outcome                                                                                                   |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Account Creation & User Management | <ul style="list-style-type: none"><li>• Create account</li><li>• Verify info</li><li>• Assign role</li><li>• Onboarding</li></ul>                       | <ul style="list-style-type: none"><li>• Role confusion</li><li>• slow admin approval</li></ul>                | <ul style="list-style-type: none"><li>• Simplify role selection</li><li>• automate verification</li></ul> |
| Sample Submission                  | <ul style="list-style-type: none"><li>• Select sample type</li><li>• Enter details</li><li>• attach documents</li><li>• confirm</li></ul>               | <ul style="list-style-type: none"><li>• Tedious data entry</li><li>• unclear barcode labeling</li></ul>       | <ul style="list-style-type: none"><li>• Simplify input</li><li>• automate barcode generation</li></ul>    |
| Order Tracking                     | <ul style="list-style-type: none"><li>• Navigate orders</li><li>• view status</li><li>• request updates</li><li>• track completion</li></ul>            | <ul style="list-style-type: none"><li>• Lack of real-time updates</li><li>• difficult filtering</li></ul>     | <ul style="list-style-type: none"><li>• Add live notifications</li><li>• improve search options</li></ul> |
| Report & Results Access            | <ul style="list-style-type: none"><li>• Choose report</li><li>• customize filters</li><li>• generate</li><li>• analyze</li><li>• share/export</li></ul> | <ul style="list-style-type: none"><li>• Overwhelming report options</li><li>• long generation times</li></ul> | <ul style="list-style-type: none"><li>• Provide templates</li><li>• optimize report generation</li></ul>  |



# User Flows

User flows visually map the steps users take to complete key tasks within the LIMS, from sample submission to report generation. These diagrams help ensure that the workflows are intuitive and efficient, minimizing friction points.

Note: Full User flow diagrams and details can be accessed in Figma:

[Link to User flows in Figma](#)

# Findings

## **Account Creation & User Management:**

Streamlining role verification and onboarding can significantly reduce friction in the user setup process.

## **Sample Submission:**

Automating barcode generation and allowing bulk submissions can improve submission accuracy and efficiency.

## **Order Tracking:**

Real-time notifications and better filtering options are crucial for user satisfaction and ensuring order visibility.

## **Report & Results Access:**

Providing customizable reports and improving report generation speed can enhance user experience, especially for researchers.

# Design Recommendations

This section presents actionable design improvements based on research insights. The recommendations focus on optimizing user experience, streamlining workflows, and improving key features like account management, sample submission, and reporting.

## **Account Creation & User Management:**

Implement a streamlined onboarding process with real-time feedback and automated role verification to reduce delays and confusion during setup.

## **Sample Submission:**

Simplify the sample submission flow by integrating bulk upload options, automating barcode generation, and ensuring users can submit orders and samples in one combined step as an alternative option.

## **Order Tracking:**

Introduce real-time order status updates with clear filtering and sorting options for better visibility into order progress.

## **Report & Results Access:**

Improve the reporting feature by allowing for customizable report templates and speeding up report generation, providing users with more flexibility and control over their data.

# Conclusion

The research provided key insights into improving the Wasatch BioLabs LIMS system:

- **Streamline workflows:** Focus on reducing steps for account creation, sample submission, and order processing to improve efficiency.
- **Enhance real-time tracking:** Integrate live status updates and better filtering options to increase visibility.
- **Improve report customization:** Allow users to create flexible reports and speed up the generation process.

These findings will guide the design phase, ensuring the system is user-friendly, scalable, and meets the needs of all users.